

Cryptography - 26_09

≡ Prove the lemma

We need to show that $1 \rightarrow 2$, $2 \rightarrow 3$ and $3 \rightarrow 1$.

$1 \rightarrow 2$) If perfect secrecy, then M and C are independent

$$Pr[M = m] = Pr[M = m|C = c] \stackrel{\text{bayes}}{=} \frac{Pr[M=m \wedge C=c]}{Pr[C=c]} = \frac{Pr[M=m] \cdot \cancel{Pr[C=c]}}{\cancel{Pr[C=c]}} = Pr[M = m]$$

.

$2 \rightarrow 3$)

Fix m from M and c from C :

$$Pr_K[Enc(K, m) = c] = Pr[\underbrace{Enc(K, M) = c}_C | M = m] = Pr[C = c | M = m] \stackrel{\text{they're independent}}{=} Pr[C = c]$$

Replacing m with m' :

$$Pr[Enc(K, m') = c] = Pr[C = c]$$

$3 \rightarrow 1$)

Take any c from C :

$$Pr[C = c] \stackrel{*_1}{=} Pr[C = c | M = m]$$

$*_1$ we claim it by property three.

If claim is true, then:

$$Pr[M = m | C = c] \cdot Pr[C = c] = Pr[M = m \wedge C = c] = Pr[C = c | M = m] \cdot Pr[M = m]$$

.

$$\implies Pr[M = m] \stackrel{*_2}{=} \frac{Pr[M=m|C=c] \cdot \cancel{Pr[C=c]}}{\cancel{Pr[C=c|M=m]}} = Pr[M = m | C = c].$$

$*_2$ By using the claim we've made before, cancelling $Pr[C = c]$ and

$$Pr[C = c | M = m].$$

It remains to prove the claim.

$$\begin{aligned} Pr[C = c] &= \sum_{m'} Pr[C = c \wedge M = m'] \stackrel{\text{bayes}}{=} \sum_{m'} Pr[C = c | M = m'] \cdot Pr[M = m'] \\ &= \sum_{m'} Pr[Enc(K, M) = c | M = m'] \cdot Pr[M = m'] = \sum_{m'} Pr[Enc(K, m') = c] \cdot Pr[M = m'] \\ &\stackrel{*_3}{=} \sum_{m'} Pr[Enc(K, m) = c] \cdot Pr[M = m'] \stackrel{*_4}{=} Pr[Enc(K, m) = c] \cdot \underbrace{\sum_{m'} Pr[M = m']}_1 \end{aligned}$$

$$= \Pr[Enc(K, m) = c] = \Pr[\underbrace{Enc(K, M)}_C = c | M = m] = \Pr[C = c | M = m]$$

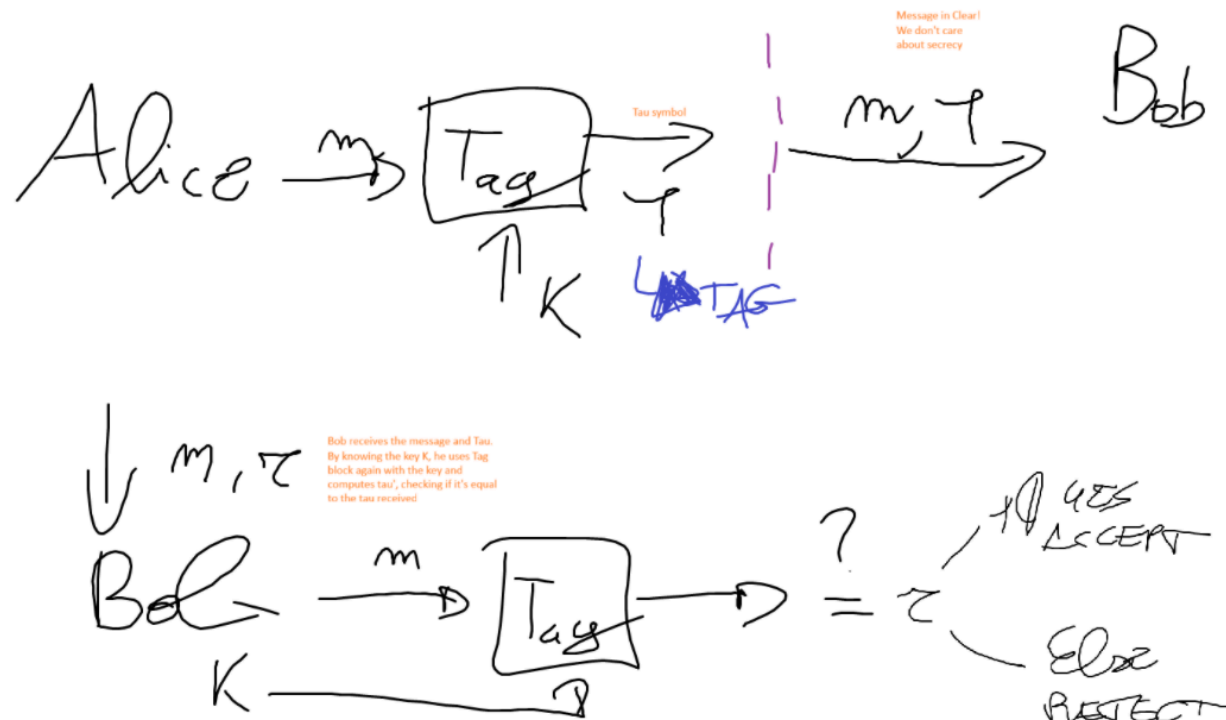
*₃ we apply property 3

*₄ because its constant, we take it out the sum

■

Message Authentication Codes

Also called MAC_s .



There is no need to define correctness here. Assuming that Tag is deterministic, then it's always correct.

Unforgeability: Should be hard to forge valid tag τ' on message m' .

If you don't know the secret key, you cannot create a valid tag for any message.

But what if you already know one?

Not only that! It should be hard to produce it even given a valid pair (m, τ) as long as $m' \neq m$.

✓ Definition (Statistical Secure Mac)

We say $\Pi = \text{Tag}$ has $\mathcal{E}$ – statistical security if

$$\forall m, m' \in \mathcal{M} \text{ with } m' \neq m$$

$\forall \tau, \tau' \in \mathcal{T}$:

$$\Pr[\text{Tag}(K, m') = \tau' | \text{Tag}(K, m) = \tau] \leq \mathcal{E}.$$

So by knowing that tau is a tag of m, we have that the probability of having tau prime as tag for m prime is less or equal than $\mathcal{E}$.

Here $\mathcal{E}$ is a parameter, as an example we could have $\mathcal{E} = 2^{-80}$.

🔗 Exercise

Impossible to get $\mathcal{E} = 0$.

Because a random $\tau' \in \mathcal{T}$ has probability $\geq \frac{1}{|\mathcal{T}|}$ to be correct.

Note that the definition is one-time! *This tells us that with a key it's ok for us to tag one message, with two it breaks.*

We will show:

- The notion is achievable.
- It's inefficient. In fact:

✓ Theorem

Any t -Time $2^{-\lambda}$ - *stat.* secure Tag has a key of size $(t + 1) \cdot \lambda$.

The definition we've seen before is one-time, we only tag one message (m' is a different story).

The size of the key grows with t .

We now show that any family of hash functions with a particular property satisfies the definition.

✓ Definition (pairwise independent)

A family $\mathcal{H} = \{h_k : \mathcal{M} \rightarrow \mathcal{T}\}_{k \in K}$ is pairwise independent if:

$\forall m, m' \in \mathcal{M}$ such that $m \neq m'$ then:

$(h(K, m), h(K, m'))$ $\ast_1$ is uniform over $\mathcal{T}^2 = \mathcal{T} \times \mathcal{T}$ for K uniform in $\mathcal{K}$ ($\mathcal{K}$ is the key space, K a key in $\mathcal{K}$).

$\ast_1$ i pick a random key from the distribution